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### Air conditioning appliance and make-up air filter tray

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#### Abstract

A single-package air conditioner unit includes a housing defining an outdoor portion and an indoor portion, an outdoor heat exchanger assembly disposed in the outdoor portion and comprising an outdoor heat exchanger and an outdoor fan, an indoor heat exchanger assembly disposed in the indoor portion and comprising an indoor heat exchanger and an indoor fan, an intake conduit extending from the housing and defining a primary air flow path upstream from the indoor heat exchanger assembly and a make-up air (MUA) air flow path, the MUA air flow bath being non-sequential with the primary air flow path, and a filter tray received within the intake conduit along the MUA air flow path, the filter tray defining a filter reception zone within the intake conduit to receive a filter cartridge therein.

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## Background/Summary

### FIELD OF THE INVENTION

(1) The present subject matter relates generally to air conditioning appliances, and more particularly to assemblies for providing make-up air to air conditioning appliances.

## BACKGROUND OF THE INVENTION

(2) Air conditioner or air conditioning appliance units are conventionally used to adjust the temperature within structures such as dwellings and office buildings. In particular, one-unit type room air conditioner units, such as single-package vertical units (SPVU), may be used to adjust the temperature in, for example, a single room or group of rooms of a structure. A typical one-unit type air conditioner or air conditioning appliance includes an indoor portion and an outdoor portion. The indoor portion generally communicates (e.g., exchanges air) with the area within a building, and the outdoor portion generally communicates (e.g., exchanges air) with the area outside a building. Accordingly, the air conditioner unit generally extends through, for example, an outer wall of the structure. Generally, a fan may be operable to rotate to motivate air through the indoor portion. Another fan may be operable to rotate to motivate air through the outdoor portion. A sealed cooling system including a compressor is generally housed within the air conditioner unit to treat (e.g., cool or heat) air as it is circulated through the indoor portion of the air conditioner unit. One or more control boards are typically provided to direct the operation of various elements of the particular air conditioner unit.

(3) Frequently, the indoor space may need to draw in air from the outdoors (i.e., make-up air). For example, if a vent fan is turned on in a bathroom or air is otherwise ejected from the indoor space, fresh air from the outdoors is required. Depending on, for example, the efficiency of the weather stripping around doors and windows, some make-up air could simply be drawn into the indoors by cracks or other openings. If such cracks are not sufficient, the flow of make-up air may be insufficient or too slow. Furthermore, government regulations, such as air flow regulations, may require that a sufficient flow of make-up air meet a minimum air flow amount. Accordingly, an air conditioner unit that can allow for the introduction of make-up air into the indoor space that meets or exceeds air flow standard amounts would be useful.

(4) As a result, it would be useful to provide an air conditioning appliance that includes features for addressing one or more of the above issues. In particular, it may be advantageous to provide an appliance or assembly with features for filtering make-up air without sacrificing air flow amounts to an air conditioning unit.

## BRIEF DESCRIPTION OF THE INVENTION

(5) Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

(6) In one exemplary aspect of the present disclosure, a single-package air conditioner unit is provided. The single-package air conditioner unit may include a housing defining an outdoor portion and an indoor portion; an outdoor heat exchanger assembly disposed in the outdoor portion and including an outdoor heat exchanger and an outdoor fan; an indoor heat exchanger assembly disposed in the indoor portion and including an indoor heat exchanger and an indoor fan; an intake conduit extending from the housing and defining a primary air flow path upstream from the indoor heat exchanger assembly and a make-up air (MUA) air flow path, the MUA air flow path being non-sequential with the primary air flow path; and a filter tray received within the intake conduit along the MUA air flow path, the filter tray defining a filter reception zone within the intake conduit to receive a filter cartridge therein.

(7) In another exemplary aspect of the present disclosure, a single-package air conditioner unit is provided. The single-package air conditioner unit may include a housing defining an outdoor portion and an indoor portion; an outdoor heat exchanger assembly disposed in the outdoor portion and including an outdoor heat exchanger and an outdoor fan; an indoor heat exchanger assembly disposed in the indoor portion and including an indoor heat exchanger and an indoor fan; an intake conduit extending from the housing and defining a primary air flow path upstream from the indoor heat exchanger assembly and a make-up air (MUA) air flow path, the MUA air flow path being non-sequential with the primary air flow path; and a filter tray removably received within the intake conduit along the MUA air flow path, the filter tray defining a filter reception zone within

the intake conduit to receive a filter cartridge therein, wherein the MUA air flow path has a minimum airflow cross-section that is less than a minimum airflow cross-section of the primary air flow path, and wherein an air flow direction of the MUA air flow path is different from an air flow direction of the primary air flow path.

(8) These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

(2) FIG. 1 provides a perspective view of an air conditioning appliance according to exemplary embodiments of the present disclosure.

(3) FIG. 2 provides a partially-transparent elevation view of the exemplary air conditioner unit of FIG. 1.

(4) FIG. 3 provides another perspective view of the exemplary air conditioner unit of FIG. 1 with a filter tray removed.

(5) FIG. 4 provides a top view of the exemplary air conditioner unit of FIG. 1 with the filter tray removed.

(6) FIG. 5 provides a front view of the exemplary air conditioner unit of FIG. 1.

(7) Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

### DETAILED DESCRIPTION

(8) Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(9) As used herein, the terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). The terms “upstream” and “downstream” refer to the relative flow direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the flow direction from which the fluid flows, and “downstream” refers to the flow direction to which the fluid flows.

(10) Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the

approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

(11) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(12) Turning now to the figures, FIGS. 1 and 2 illustrate an exemplary air conditioner appliance (e.g., air conditioner **100**). As shown, air conditioner **100** may be provided as a one-unit type air conditioner **100**, such as a single-package vertical unit. Air conditioner **100** includes a package housing **114** supporting an indoor portion **112** and an outdoor portion **110**. Generally, air conditioner **100** defines a vertical direction V, lateral direction L, and transverse direction T. Each direction V, L, T is perpendicular to each other, such that an orthogonal coordinate system is generally defined.

(13) In some embodiments, housing **114** contains various other components of the air conditioner **100**. Housing **114** may include, for example, a rear opening **116** (e.g., with or without a grill or grate thereacross) and a front opening **118** (e.g., with or without a grill or grate thereacross) may be spaced apart from each other along the transverse direction T. The rear opening **116** may be part of the outdoor portion **110**, while the front opening **118** is part of the indoor portion **112**. Components of the outdoor portion **110**, such as an outdoor heat exchanger **120**, outdoor fan **124**, and compressor **126** may be enclosed within housing **114** between front opening **118** and rear opening **116**. In certain embodiments, one or more components of outdoor portion **110** are mounted on a base pan **136**, as shown.

(14) During certain operations, air may be drawn to outdoor portion **110** through rear opening **116**. Specifically, an outdoor inlet **128** defined through housing **114** may receive outdoor air motivated by outdoor fan **124**. Within housing **114**, the received outdoor air may be motivated through or across outdoor fan **124**. Moreover, at least a portion of the outdoor air may be motivated through or across outdoor heat exchanger **120** before exiting the rear opening **116** at an outdoor outlet **130**. It is noted that although outdoor inlet **128** is illustrated as being defined above outdoor outlet **130**, alternative embodiments may reverse this relative orientation (e.g., such that outdoor inlet **128** is defined below outdoor outlet **130**) or provide outdoor inlet **128** beside outdoor outlet **130** in a side-by-side orientation, or another suitable discrete orientation.

(15) As shown, indoor portion **112** may include an indoor heat exchanger **122**, a blower fan **142**, and a heating unit **132**. These components may, for example, be housed behind the front opening **118**. A bulkhead **134** may generally support or house various other components or portions thereof of the indoor portion **112**, such as the blower fan **142**. Bulkhead **134** may generally separate and define the indoor portion **112** and outdoor portion **110** within housing **114**. Additionally or alternatively, bulkhead **134** or indoor heat exchanger **122** may be mounted on base pan **136** (e.g., at a higher vertical position than outdoor heat exchanger **120**), as shown.

(16) During certain operations, air may be drawn to indoor portion **112** through front opening **118**. Specifically, an indoor inlet **138** defined through housing **114** may receive indoor air motivated by

blower fan **142**. At least a portion of the indoor air may be motivated through or across indoor heat exchanger **122** (e.g., before passing to bulkhead **134**). From blower fan **142**, indoor air may be motivated (e.g., across heating unit **132**) and returned to the indoor area of the room through an indoor outlet **140** defined through housing **114** (e.g., above indoor inlet **138** along the vertical direction V). Optionally, one or more conduits (not pictured) may be mounted on or downstream from indoor outlet **140** to further guide air from air conditioner **100**. It is noted that although indoor outlet **140** is illustrated as generally directing air upward, it is understood that indoor outlet **140** may be defined in alternative embodiments to direct air in any other suitable direction.

(17) Outdoor and indoor heat exchanger **120, 122** may be components of a thermodynamic assembly (i.e., sealed system), which may be operated as a refrigeration assembly (and thus perform a refrigeration cycle) or, in the case of the heat pump unit embodiment, a heat pump (and thus perform a heat pump cycle). Thus, as is understood, exemplary heat pump unit embodiments may be selectively operated perform a refrigeration cycle at certain instances (e.g., while in a cooling mode) and a heat pump cycle at other instances (e.g., while in a heating mode). By contrast, exemplary A/C exclusive unit embodiments may be unable to perform a heat pump cycle (e.g., while in the heating mode), but still perform a refrigeration cycle (e.g., while in a cooling mode).

(18) The sealed system may, for example, further include compressor **126** (e.g., mounted on base pan **136**) and an expansion device (e.g., expansion valve or capillary tube—not pictured), both of which may be in fluid communication with the heat exchangers **120, 122** to flow refrigerant therethrough, as is generally understood. The outdoor and indoor heat exchanger **120, 122** may each include coils **146, 148**, as illustrated, through which a refrigerant may flow for heat exchange purposes, as is generally understood.

(19) It is noted that although a sealed system is described above (e.g., as a thermodynamic assembly), one of ordinary skill in the art would, in light of the present disclosure, understand that such a sealed system may be substituted for other suitable heat-exchange systems, such as a system relying on shape-memory alloys (SMA). For instance, a pair of discrete fluid circuits (e.g., a hot circuit and a cold circuit) each having a discrete volume of heat-carrying fluid (e.g., water, brine, glycol, air, etc.) may be separately connected to a compression unit—the compression unit housing a plurality of plate stacks each having one or more plates formed from one or more SMA material (e.g., copper-nickel-aluminum or nickel-titanium). Separate heat exchangers may generally be provided on the circuits in place of the evaporator and the condenser of a sealed system. In particular, a first heat exchanger may be provided on the cold circuit (e.g., in place of the evaporator) to absorb heat from the adjacent air and impart such absorbed heat to the heat-carrying fluid within the cold circuit. Thus, the first heat exchanger may also be referred to as an “evaporator” herein. Similarly, a second heat exchanger may be provided on the hot circuit (e.g., in place of the condenser) to release heat to the adjacent air from the heat-carrying fluid within the hot circuit. Thus, the second heat exchanger may also be referred to as a “condenser” herein.

(20) The compression unit may facilitate or direct heat between the circuits. As an example, the compression unit may have four discrete plate stacks, each being separately compressed or released by a corresponding mechanical press or vice (e.g., hydraulic ram or electric actuator). During use, the plate stacks may be compressed and released (e.g., alternated between a compressed state or stroke and a released state or stroke) separately such that at any given moment one plate stack is compressed, one plate stack is released, one plate stack is mid-compression, and one plate stack is mid-release. Heat-carrying fluid in the cold circuit may flow through the first heat exchanger, before being directed (e.g., by a series of valves or pumps) into the plate stack that is currently compressed. The compressed plate stack may then be moved to the released state, in turn absorbing heat from the heat-carrying fluid before the heat-carrying fluid within the now-released plate stack is returned to the cold circuit (e.g., to repeat the cycle). In contrast to the cold circuit, heat-carrying fluid in the hot circuit may flow through the second heat exchanger and be directed (e.g., by a

separate series of valves or pump) into the plate stack that is currently released. The released plate stack may then be compressed (i.e., moved to the compressed state), in turn releasing heat from the plate stack to the heat-carrying fluid before the heat-carrying fluid within the now-compressed plate stack is returned to the hot circuit (e.g., to repeat the cycle). The use of four plate stacks may allow both circuits to run continuously.

(21) A plenum **166** may be provided to direct air to or from housing **114**. When installed, plenum **166** may be selectively attached to (e.g., fixed to or mounted against) housing **114** (e.g., via a suitable mechanical fastener, adhesive, gasket, etc.) and extend through a structure wall **150** (e.g., an outer wall of the structure within which air conditioner **100** is installed). For instance, plenum **166** may extend (e.g., parallel to the transverse direction T) through a hole or channel in the structure wall **150** that passes from an internal surface **154** to an external surface **156**.

(22) As will be described in greater detail below, a make-up air assembly **200** may be provided to selectively direct outdoor or make-up air to the indoor portion **112**. Specifically, make-up air assembly **200** may direct outdoor air through the structure outer or wall **150** of the structure within which air conditioner **100** is installed (e.g., via plenum **166**) and to indoor heat exchanger **122** without first directing such outdoor or make-up air through housing **114**. To that end, make-up air assembly **200** may include one or more air ducts or conduits (e.g., intake conduit **210** or secondary air duct **212**) defining one or more air paths outside of housing **114**. During use, the flow of make-up air may thus be fluidly isolated from the flow of air through outdoor portion **110**.

(23) The operation of air conditioner **100** including compressor **126** (and thus the sealed system generally), blower fan **142**, outdoor fan **124**, heating unit **132**, and other suitable components may be controlled by a control board or controller **158**. Controller **158** may be in communication (via for example a suitable wired or wireless connection) to such components of the air conditioner **100**. By way of example, the controller **158** may include a memory and one or more processing devices such as microprocessors, CPUs or the like, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with operation of air conditioner **100**. The memory may be a separate component from the processor or may be included onboard within the processor. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH.

(24) Air conditioner **100** may additionally include a control panel **160** and one or more user inputs **162**, which may be included in control panel **160**. The user inputs **162** may be in communication with the controller **158**. A user of the air conditioner **100** may interact with the user inputs **162** to operate the air conditioner **100**, and user commands may be transmitted between the user inputs **162** and controller **158** to facilitate operation of the air conditioner **100** based on such user commands. A display **164** may additionally be provided in the control panel **160**, and may be in communication with the controller **158**. Display **164** may, for example be a touchscreen or other text-readable display screen, or alternatively may simply be a light that can be activated and deactivated as required to provide an indication of, for example, an event or setting for the air conditioner **100**.

(25) As noted above, make-up air assembly **200** may be generally provided to selectively direct outdoor air to the indoor portion **112**. To that end, make-up air assembly **200** may include an intake conduit **210** that defines an intake passage **214** upstream from indoor inlet **138**. As shown, intake conduit **210** extends outward from housing **114**. For instance, intake passage **214** may extend along a passage axis X (e.g., horizontal or parallel to the transverse direction T), which the intake conduit **210** generally surrounds or radially bounds. In some such embodiments, intake passage **214** is parallel to passage axis X. When assembled, intake conduit **210** may be mounted to housing **114**, such as on an outer surface **230** of housing **114**. In turn, intake passage **214** may extend from a primary air inlet **216** (i.e., primary inlet), which is defined as an opening or aperture of intake conduit **210**, to indoor inlet **138**. Thus, primary air inlet **216** is spaced apart from indoor inlet **138** (e.g., along the transverse direction T). In some embodiments, primary air inlet **216** is coaxial with

indoor inlet **138**. For instance, both primary air inlet **216** and indoor inlet **138** may be defined along the passage axis X. In turn, intake passage **214** may be a linear passage from primary air inlet **216** to indoor inlet **138**.

(26) Generally, primary air inlet **216** defines an airflow cross section (e.g., minimum cross section) along a plane perpendicular to airflow through primary air inlet **216**. For instance, in the illustrated embodiments, the airflow cross section of primary air inlet **216** is defined by the dimensions of a height multiplied by a width thereof.

(27) Along with defining primary air inlet **216**, intake conduit **210** may define a secondary air inlet **218** (i.e., secondary inlet). In particular, secondary air inlet **218** may be defined separate from primary air inlet **216**. When assembled, secondary air inlet **218** may be spaced apart from primary air inlet **216**. For instance, secondary air inlet **218** may be defined in fluid parallel (e.g., non-sequential) to primary air inlet **216**. Thus, airflow through secondary air inlet **218** to intake passage **214** may be distinct from airflow through primary air inlet **216**. Moreover, upstream from intake passage **214**, the airflows through secondary air inlet **218** and primary air inlet **216** may be independent from (i.e., not commingled with) each other.

(28) In some embodiments, secondary air inlet **218** is defined along a non-parallel angle relative to primary air inlet **216** (i.e., such that primary air inlet **216** and secondary air inlet **218** are not defined along geometric parallel axes). For instance, secondary air inlet **218** may be defined through intake conduit **210** perpendicular to primary air inlet **216** (e.g., perpendicular to passage axis X). In optional embodiments, secondary air inlet **218** is defined above primary air inlet **216**. Thus, airflow through secondary air inlet **218** to intake passage **214** may flow downward. In additional or alternative embodiments, secondary air inlet **218** is closer to indoor inlet **138** (e.g., relative to the passage axis X) than primary air inlet **216**. Thus, secondary air inlet **218** may be proximal to indoor inlet **138** while primary air inlet **216** is distal to indoor inlet **138**.

(29) Generally, secondary air inlet **218** defines an airflow cross section (e.g., minimum cross section) along a plane perpendicular to airflow through secondary air inlet **218**. For instance, in the illustrated embodiments, the airflow cross section of secondary air inlet **218** is defined by the dimensions of a length multiplied by a width thereof. In certain embodiments, the airflow cross section of secondary air inlet **218** is less than the airflow cross section of primary air inlet **216**.

(30) As shown, a filter panel **220** may be disposed (e.g., selectively or removably disposed) on intake conduit **210**. In particular, filter panel **220** may be disposed in fluid communication with intake passage **214** to filter air thereto. For instance, filter panel **220** may be in fluid communication with primary air inlet **216** while being spaced apart from secondary air inlet **218**. During use, airflow to intake passage **214** through primary air inlet **216** may thus be forced through filter panel **220** in order to flow to intake passage **214**. By contrast, airflow to intake passage **214** through secondary air inlet **218** may advantageously bypass filter panel **220** altogether. Optionally, indoor inlet **138** may be unobstructed by any filtration media, ensuring a direct flow path from intake passage **214** to the indoor portion **112**. Notably, bypassing filter panel **220** may prevent significant resistance to make-up air (e.g., while ensuring filtration of most of the airflow, such as the non-makeup airflow to indoor inlet **138**).

(31) In some embodiments, filter panel **220** is disposed in front primary air inlet **216** (e.g., along the transverse direction T or otherwise outside from intake passage **214**). Moreover, filter panel **220** may be upstream from primary air inlet **216**. One or more mounting brackets **222** may be provided to hold filter panel **220** on intake conduit **210**. For instance, as illustrated, a pair of mounting brackets **222** that each defining a discrete support channel to slidably receive filter panel **220** may be provided on opposite ends (e.g., opposite lateral ends or vertical ends) of intake conduit **210** or primary air inlet **216**. As shown, each mounting bracket **222** may be opened at one end (e.g., a top end) while being closed at an opposite end (e.g., a bottom end) to support filter panel **220** or otherwise prevent filter panel **220** from sliding directly through (i.e., out of) the mounting brackets **222** during installation of filter panel **220** on intake conduit **210**. Filter panel **220** itself may be



provided as any suitable frame or structure including a suitable air filtration media (e.g., cellulose, fiberglass, foam, etc.).

(32) In some embodiments, a secondary air duct **212** is mounted or attached to intake conduit **210** to direct outdoor (i.e., make-up) air to secondary air inlet **218**. Thus, secondary air duct **212** may be disposed upstream from secondary air inlet **218** to direct air thereto. Moreover, secondary air duct **212** may define a secondary passage **224** that extends from an outdoor end **226** to an indoor end **228**. In certain embodiments, outdoor end **226** is positioned at or proximal to plenum **166** while indoor end **228** is positioned at or proximal to secondary air inlet **218**, as shown.

(33) When assembled, secondary air duct **212** may extend from intake conduit **210** outside of housing **114**. Thus, secondary passage **224** may be defined outside of housing **114** apart from indoor portion **112** and outdoor portion **110**. In certain embodiments, secondary air duct **212** is disposed on outer surface **230** of housing **114**. Thus, while secondary air duct **212** is separate from housing **114**, secondary air duct **212** may be held to housing **114** (e.g., as a single unit) without commingling air through outdoor portion **110** and secondary passage **224**, or without motivating air within housing **114** across an exterior surface of secondary air duct **212** (e.g., opposite from the interior secondary passage **224**). Optionally, an insulation layer (e.g., insulating foam, sheet, or panels) may be disposed on the exterior surface of secondary air duct **212** outside of housing **114**.

(34) Outdoor end **226** of secondary air duct **212** defines a duct intake **232** through which outdoor or make-up air may enter secondary passage **224**. In certain embodiments, duct intake **232** may be disposed directly above the outdoor outlet **130** (e.g., at the plenum **166**). In order to direct air from an outdoor region to intake conduit **210**, duct intake **232** is generally offset or spaced apart from secondary air inlet **218** in or along at least one direction (e.g., the transverse direction T). As shown, duct intake **232** may further be offset from secondary air inlet **218** in a second direction (e.g., in the vertical direction V or lateral direction L). In the illustrated embodiments, indoor end **228** (and thus secondary air inlet **218**) is both horizontally and vertically offset from duct intake **232**. For instance, secondary air inlet **218** may be transversely offset and lower than the duct intake **232**. Airflow through secondary passage **224** may thus be drawn forward and downward, notably maintaining a compact assembly while minimizing the resistance of airflow through secondary passage **224**.

(35) Referring now generally to FIGS. 3 through 5, MUA assembly **200** may include a filter tray **240**. As described above, air from the outdoor region (make-up air, MUA) may be directed into intake conduit **210** via secondary air duct **212** (e.g., through secondary air inlet **218**). Further, the MUA may be introduced into intake conduit **210** downstream from filter panel **220** (e.g., main flow or primary flow filter panel **220**). Accordingly, filter tray **240** may be accommodated within intake conduit **210**. In detail, filter tray **240** may be removably received within intake conduit **210** along a MUA air flow path (e.g., defined through secondary air duct **212** and intake conduit **210**). Thus, a secondary or MUA filter cartridge **258** may be selectively accommodated within filter tray **240**. For instance, filter tray **240** may define a filter reception zone **242** within intake conduit **210**.

Secondary filter cartridge **258** may be provided within filter reception zone **242**.

(36) Filter tray **240** may include a main body panel **244**. Main body panel **244** may include a front panel **246** and a pair of side panels **248** extending from front panel **246**. Front panel **246** may be defined primarily along the vertical direction V and the lateral direction L. For instance, front panel **246** may define a width **2461** along the lateral direction L and a height **2462** along the vertical direction V. According to at least some embodiments, width **2461** is greater than height **2462**. For instance, width **2461** may be about equal to a width along the lateral direction L of intake conduit **210**. Thus, a width of filter reception zone **242** may be equal to width **2461** of front panel **246**, and therefore equal to the width of intake conduit **210**. Advantageously, the MUA air flow supplied to intake conduit **210** may be thoroughly filtered (e.g., by a MUA filter cartridge) before being introduced to indoor heat exchanger **122** and subsequently supplied to the indoor area.

(37) The pair of side panel **248** may include first and second side panels **248**. Each of the first and

second side panels **248** may extend from front panel **246** (e.g., along the transverse direction T). Each of the first and second side panels **248** may be similarly provided (and mirrored about a plane defined along the vertical direction V and transverse direction T). Accordingly, a single side panel **248** will be described herein with the understanding that the description may apply to each of the pair of side panels **248**.

(38) Side panel **248** may extend a predetermined distance along the transverse direction T. For instance, side panel **248** may extend at a right angle with respect to front panel **246**. Moreover, side panel **248** may be provided at or near a lateral edge of front panel **246**. In detail, an extending length **2481** of side panel **246** may be approximately equal to a depth (along the transverse direction T) of intake conduit **210**. Accordingly, a distal edge of side panel **248** may contact an inner surface **211** of intake conduit **210** (e.g., an exterior surface of housing **114**). Moreover, filter reception zone **242** may be defined at least in part by extending length **2481** of side panel **248** and width **2461** of front panel **246**. Advantageously, a full cross-section (e.g., along the vertical direction V) of the MUA air flow path may be within filter reception zone **242**.

(39) Filter tray **240** may include a plurality of flanges **250** extending from main body **244**. For instance, the plurality of flanges **250** may extend toward filter reception zone **242** (e.g., along the lateral direction L and transverse direction T). In detail, the plurality of flanges **250** may include a front flange **252** and a pair of side flanges **254**. Front flange **252** may extend from front panel **246**. In detail, front flange **252** may extend at a right angle with respect to front panel **246**. Front flange **252** may thus extend rearward (e.g., along the transverse direction T toward housing **114**) from front panel **246**. Additionally or alternatively, front flange **252** may extend from a bottom (e.g., along the vertical direction V) of front panel **246**. Moreover, front flange **252** may extend to a predetermined distance from front panel **246**. According to some embodiments, the extending length **2521** (e.g., along the transverse direction T) of front flange **252** is between about 10% and about 20% of extending length **2481** of side panel **248**.

(40) Similarly, the pair of side flanges **254** may extend from the pair of side panels **248**, respectively. For instance, a first side flange **254** may extend toward filter reception zone **242** from a first side panel **248** while a second side flange **254** extends toward filter reception zone **242** from a second side panel **248**. Additionally or alternatively, the pair of side flanges **254** may extend toward each other (e.g., along the lateral direction L) into filter reception zone **242**. Hereinafter, a single side flange **254** will be described in detail with the understanding that the description applies to each side flange **254**. Similar to front flange **252**, side flange **254** may extend from a bottom (e.g., along the vertical direction V) of side panel **248**. Moreover, side flange **254** may extend to a predetermined distance from side panel **248**. According to some embodiments, the extending length **2541** (e.g., along the transverse direction T) of side flange **254** is between about 5% and about 10% of width **2461** of front panel **246**. Collectively, front flange **252** and side flanges **254** may define a tray aperture **256** to permit air therethrough. In detail, tray aperture **256** may be defined along the lateral direction L between the pair of side flanges **254** and along the transverse direction T between front flange **252** and inner surface **211** of intake conduit **210**.

(41) A filter cartridge (e.g., MUA filter cartridge) **258** may be selectively supported by the plurality of flanges **250** (e.g., along the vertical direction V). In detail, filter cartridge **258** may rest on the plurality of flanges **250** within filter reception zone **242**. An exemplary filter cartridge **258** is shown in FIG. 5. As shown in FIG. 5, filter cartridge **258** may be provided at or near a bottom (e.g., along the vertical direction V) of filter reception zone **242**. For instance, filter cartridge **258** may occupy a certain percentage of space defined by filter tray **240** (e.g., three-dimensional space). For one example, a height **2581** of filter cartridge **258** is less than the height **2461** of front panel **246**. Accordingly, a dispersion gap **260** may be formed therein.

(42) Dispersion gap **260** may be defined within the MUA air flow path. According to some embodiments, dispersion gap **260** is provided above (e.g., along the vertical direction V) filter reception zone **242** (or above filter cartridge **258**). Dispersion gap **260** may define a space between,

for instance, secondary air duct (or MUA duct) **212** and filter reception zone **242**. In detail, dispersion gap **260** may form an enlarged space above filter reception zone **242** for the make-up air to expand before entering filter cartridge **258**. According to at least one example, a width (e.g., along the lateral direction L) of dispersion gap **260** is equal to a width (along the lateral direction L) of filter reception zone **242**. Additionally or alternatively, a height (e.g. along the vertical direction V) of dispersion gap **260** may be greater than a height (along the vertical direction V) of filter reception zone **242**. However, it should be noted that the relative heights of each of dispersion gap **260** and filter reception zone **242** may vary according to specific filter cartridges provided within filter reception zone **242**.

(43) MUA duct **212** (secondary air duct) may define a first cross-sectional area. For instance, MUA duct may define a width **262** along the lateral direction L and a length **264** along the transverse direction T. For purposes of this example, the cross-sectional area is determined at secondary air inlet **218**. As shown in the FIGS., the cross-sectional area of MUA duct **212** is less than a cross-sectional area of dispersion gap **260**. Accordingly, outside air supplied to intake conduit **210** via MUA duct **212** may expand within dispersion gap **260** before entering, e.g., filter cartridge **258**. Advantageously, the outside (e.g., make-up) air may easily flow through filter cartridge **258** before being supplied to indoor heat exchanger **122**. Accordingly, the flow rate of the make-up air may be maximized.

(44) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

## Claims

1. A single-package air conditioner unit defining a vertical direction, a lateral direction, and a transverse direction, the single-package air conditioner unit comprising: a housing comprising an outdoor portion and an indoor portion, the indoor portion defining an indoor inlet for receipt of indoor air; an outdoor heat exchanger assembly disposed in the outdoor portion and comprising an outdoor heat exchanger and an outdoor fan; an indoor heat exchanger assembly disposed in the indoor portion and comprising an indoor heat exchanger and an indoor fan, the indoor heat exchanger being in downstream fluid communication with the indoor inlet; an intake conduit mounted to an outer surface of the housing, the intake conduit defining an intake passage upstream from the indoor inlet, the intake conduit further defining a primary air inlet upstream of the intake passage and a secondary air inlet upstream of the intake passage, the secondary air inlet being non-sequential with the primary air inlet; a make-up air (MUA) duct for directing outdoor air to the indoor portion, the MUA duct in upstream fluid communication with the secondary air inlet, the MUA duct and a portion of the intake conduit defining a MUA air flow path upstream of the intake passage; and a filter tray received within the intake conduit along the MUA air flow path, the filter tray defining a filter reception zone within the intake conduit to receive a filter cartridge therein, the filter reception zone being in downstream fluid communication with the secondary air inlet and upstream fluid communication with the intake passage wherein the primary air inlet is coaxial with the indoor inlet, and wherein the secondary air inlet is perpendicular to the primary air inlet and positioned above the primary air inlet.

2. The single-package air conditioner unit of claim 1, wherein the filter tray further defines a dispersion gap, the dispersion gap being provided within the intake conduit along the MUA air

flow path upstream from the filter reception zone.

3. The single-package air conditioner unit of claim 2, wherein a width of the dispersion gap along the lateral direction is equal to a width of the filter reception zone.

4. The single-package air conditioner unit of claim 2, wherein a height of the dispersion gap along the vertical direction is greater than a height of the filter reception zone.

5. The single-package air conditioner unit of claim 2, wherein the MUA duct is disposed on the outer surface of the housing, and wherein the MUA duct is in upstream fluid communication with the dispersion gap to direct air thereto.

6. The single-package air conditioner unit of claim 5, wherein a cross-sectional area of the MUA duct along the MUA air flow path is less than a cross-sectional area of the dispersion gap along the MUA air flow path.

7. The single-package air conditioner unit of claim 1, wherein the filter tray comprises: a main body panel; and a plurality of flanges extending from the main body panel toward the housing, the plurality of flanges defining a tray aperture to permit air therethrough, wherein the plurality of flanges selectively support the filter cartridge thereon within the filter reception zone.

8. The single-package air conditioner unit of claim 7, wherein the main body panel comprises: a front panel extending along the lateral direction; and a pair of side panels extending from the front panel along the transverse direction from opposite lateral ends of the front panel toward the housing.

9. The single-package air conditioner unit of claim 8, wherein the plurality of flanges comprises: a front flange extending toward the housing from a bottom of the front panel; and a pair of side flanges extending toward the housing from a bottom of the pair of side panels.

10. The single-package air conditioner unit of claim 1, further comprising a filter panel disposed on the intake conduit in upstream fluid communication with the intake passage apart from the MUA air flow path.

11. The single-package air conditioner unit of claim 10, wherein the filter panel is disposed in front of the primary air inlet and in fluid communication with the primary air inlet, and wherein the filter panel is spaced apart from the secondary air inlet.

12. A single-package air conditioner unit defining a vertical direction, a lateral direction, and a transverse direction, the single-package air conditioner unit comprising: a housing comprising an outdoor portion and an indoor portion, the indoor portion defining an indoor inlet for receipt of indoor air; an outdoor heat exchanger assembly disposed in the outdoor portion and comprising an outdoor heat exchanger and an outdoor fan; an indoor heat exchanger assembly disposed in the indoor portion and comprising an indoor heat exchanger and an indoor fan, the indoor heat exchanger being in downstream fluid communication with the indoor inlet; an intake conduit mounted to an outer surface of the housing, the intake conduit defining an intake passage upstream from the indoor inlet, the intake conduit further defining a primary air inlet upstream of the intake passage and a secondary air inlet upstream of the intake passage, the secondary air inlet being non-sequential with the primary air inlet; a make-up air (MUA) duct for directing outdoor air to the indoor portion, the MUA duct in upstream fluid communication with the secondary air inlet, the MUA duct and a portion of the intake conduit defining a MUA air flow path upstream of the intake passage; a filter tray removably received within the intake conduit along the MUA air flow path, the filter tray defining a filter reception zone within the intake conduit to receive a filter cartridge therein, the filter reception zone being in downstream fluid communication with the secondary air inlet and upstream fluid communication with the intake passage, wherein the secondary air inlet has a minimum airflow cross-section that is less than a minimum airflow cross-section of the primary air inlet, and wherein an air flow direction of the secondary air inlet is different from an air flow direction of the primary air inlet; and a filter panel disposed on the intake conduit in fluid communication with the intake passage apart from the MUA air flow path, wherein the filter panel is disposed in front of the primary air inlet and in fluid communication with the primary air inlet,

and wherein the filter panel is spaced apart from the secondary air inlet.

13. The single-package air conditioner unit of claim 12, wherein the filter tray further defines a dispersion gap, the dispersion gap being provided within the intake conduit along the MUA air flow path upstream from the filter reception zone.

14. The single-package air conditioner unit of claim 13, wherein a width of the dispersion gap along the lateral direction is equal to a width of the filter reception zone.

15. The single-package air conditioner unit of claim 13, wherein a height of the dispersion gap along the vertical direction is greater than a height of the filter reception zone.

16. The single-package air conditioner unit of claim 12, wherein the filter tray comprises: a main body panel; and a plurality of flanges extending from the main body panel toward the housing, the plurality of flanges defining a tray aperture to permit air therethrough, wherein the plurality of flanges selectively supports the filter cartridge thereon within the filter reception zone.

17. The single-package air conditioner unit of claim 16, wherein the main body panel comprises: a front panel extending along the lateral direction; and a pair of side panels extending from the front panel along the transverse direction from opposite lateral ends of the front panel toward the housing.

18. The single-package air conditioner unit of claim 17, wherein the plurality of flanges comprises: a front flange extending toward the housing from a bottom of the front panel; and a pair of side flanges extending toward the housing from a bottom of the pair of side panels.

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